

Runbook: Terraform Operations

NimbusCloud Platform — Platform Engineering Team

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Standard Terraform Workflow

```
cd infrastructure/terraform

# 1. Always check state before making changes
terraform init
terraform plan

# 2. Review the plan carefully – look for unexpected destroy/replace
# 3. Apply only if plan looks correct
terraform apply

# 4. If applying to production – always use -target first to scope changes
terraform apply -target=aws_s3_bucket.assets
```

Current State Issues

Running `terraform plan` will show errors:

1. **Missing variables** — `environment` and `bucket_suffix` not in `terraform.tfvars`
2. **Wrong output reference** — `outputs.tf` references `aws_lb.nimbuscloud_alb` which was renamed to `aws_lb.main`

Fix both before running `apply`.

State Drift

If AWS resources were changed outside Terraform (console or CLI), state drift occurs.

```
# See what Terraform thinks vs what exists
terraform plan

# Import an existing resource into state (if it was created outside Terraform)
terraform import aws_s3_bucket.assets nimbuscloud-platform-assets-prod-123456789012

# Remove a resource from state without destroying it (if intentionally unmanaged)
terraform state rm aws_resource.name
```

Destroy (emergency only)

```
# DANGEROUS – destroys all managed resources
# Get approval from Priya before running
terraform destroy

# Safer: destroy a specific resource
terraform destroy -target=aws_instance.name
```